



The Impact of the Omega-6 : Omega-3 Balance on Oncological Therapy

Scientific Evidence for Optimization of Outcomes in Chemotherapy

Presentation for a Panel of Oncologists

Based on Randomized Clinical Trials, Meta-Analyses and Cohort Studies

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The Omega-6 : Omega-3 Ratio

OMEGA-6 (Arachidonic Acid)

Precursor of PRO-inflammatory eicosanoids:

Prostaglandin E₂ (PGE₂)

Leukotriene B₄ (LTB₄)

Thromboxane A₂ (TXA₂)

Activates NF-κB → tumor proliferation

Increases COX-2 → angiogenesis

OMEGA-3 (EPA + DHA)

Precursor of ANTI-inflammatory mediators:

Resolvins (RvE1, RvD1)

Protectins (PD1/NPD1)

Maresins (MaR1)

Inhibits NF-κB → tumor apoptosis

Reduces COX-2 → anti-angiogenic

The RATIO determines the balance between inflammation and resolution — a critical factor in response to oncological treatment

The Evolution of the Ratio in the Modern Diet



Paleolithic



Trad. Japan



Western Diet



Oncol. Pt.

Simopoulos, 2002: Humans evolved with ~1:1. The Western diet presents 15:1 to 16.7:1.

A ratio of 2.5:1 reduced rectal cell proliferation in colorectal cancer patients.

Lower ratio in women with breast cancer was associated with lower risk.

Oncological patients frequently present ratios >20:1 due to tumor metabolic alterations.

The Oncological Patient and Systemic Inflammation

TUMOR

Produces pro-inflammatory
cytokines
(TNF- α , IL-6, IL-1 β)

CHEMOTHERAPY

Generates ROS
Immunosuppression
Neurotoxicity

HIGH RATIO ω 6: ω 3

Amplifies inflammatory
cascade via PGE₂
Promotes resistance

The triad Tumor + Chemotherapy + High Ratio creates a vicious cycle of chronic inflammation that compromises therapeutic efficacy, accelerates cachexia and increases adverse effects.



RATIO $\geq$ 15:1

Negative Impact on Oncological Therapy

Evidence across 4 critical domains

NF-κB / COX-2 Pathway: Tumor Aggressiveness

Molecular Mechanisms

ω-6 elevated → ↑ AA in membranes
↑ COX-2 → ↑ PGE₂ → proliferation, angiogenesis
↑ NF-κB → pro-tumor genes (YY1, VEGF-A, TGF-β)
↑ P-gp and MRP1 → chemoresistance
↑ OXLAMs → systemic inflammatory activation

Experimental Evidence

PMC9181584 (2022) — Lung Cancer:

ω-6-rich diet: ↑ NF-κB p65, YY1, COX-2, VEGF-A. Greater tumor dedifferentiation.

PMC4773771 — Multi-Cancer:

Cells with high ratio maintain ↑ GSH → resistance to chemotherapy.

PMC6269634 — Inflammation:

OXLAMs activate NF-κB, ↑ cytokines and adhesion molecules.

Ratio 15-16.7:1 → more aggressive tumor phenotype, with greater angiogenesis and therapeutic resistance

UK Biobank: High Ratio and Mortality

n = 85,425 participants | 6,461 deaths | PMC9882493

+26%

Total
Mortality

+14%

Cancer
Mortality

+31%

Cardiovascular
Mortality

Highest vs. lowest quintile of plasma ω -6/ ω -3 ratio. All $P_{trend} < 0.05$

Mediated via CRP (inflammation), SHBG, testosterone and IGF-1 (cancer). Dose-dependent effects.

Cancer Cachexia and Chemoresistance

Cachexia (50-80% of patients)

TNF- α , IL-6 $\rightarrow$ muscle degradation
Ubiquitin-proteasome pathway activated
Excessive lipolysis (ZAG)
Insulin resistance
High ratio amplifies all of this

Chemoresistance

$\uparrow$ P-gp/MRP1 $\rightarrow$ drug efflux
 $\uparrow$ GSH $\rightarrow$ protection against ROS
 $\uparrow$ NF- κ B $\rightarrow$ Bcl-2, survivin
CSCs protected
Sarcopenia $\rightarrow$ $\downarrow$ CT tolerance

Consequences: $\downarrow$ response rate • $\uparrow$ dose reductions • $\uparrow$ interruptions • $\downarrow$ overall survival • $\downarrow$ quality of life



WHAT IF WE REDUCE THE RATIO?

Ratio $\leq$ 3:1 — Clinical Evidence of Benefits

Chemosensitization: Molecular Mechanisms

Cell Membranes

EPA/DHA incorporate into phospholipids → ↑ permeability
→ ↑ drug absorption. ↓ P-gp and MRP1.

Selective Oxidative Stress

↑ SELECTIVE lipid peroxidation in tumor cells. ↓ Tumor GSH. Normal cells protected.

Signaling Pathways

NF-κB inhibition → ↓ Bcl-2. ↓ Wnt/β-catenin. ↓ ERK/AKT
→ ↑ apoptosis. Notch/Hedgehog modulation.

Cancer Stem Cells (CSCs)

EPA/DHA induce apoptosis in colon CSCs (CD133+) and breast. ↓ Self-renewal. ↓ Recurrence.

Response Rate: Lung Cancer (NSCLC)

RCT | n=46 | Carboplatin + Vinorelbine/Gemcitabine ± 2.5g EPA+DHA/day

Without Fish Oil

25.8%

Response Rate

P = 0.008

With Fish Oil

60.0%

Response Rate

Clinical Benefit: 80% vs 41.9% (P=0.02) | 1-year Survival: 60% vs 38.7%

Tumor Progression: Colorectal Cancer

RCT | n=30 | CRC Stage II-IV | 2g fish oil/day × 9 weeks | Nutr Cancer 2016

330

days to progression
(± 135) — Control

+80%

P = 0.04

593

days to progression
(± 211) — Fish Oil

Bognoux 2009 — Metastatic Breast: 1.8g DHA + anthracyclines. High DHA incorporation → ↑ time to progression and survival.

CALGB 89803 — CRC Stage III (n=1264): Highest quartile marine ω-3 → HR 0.72 (0.54-0.97) for DFS. Stronger effect with high tumor COX-2.

Anti-Cachexia: Body Mass Preservation

NSCLC
(Murphy 2011)

+1.1 kg

vs. -2.3 kg without ω -3

Muscle preservation

n=40 | Cancer 117:1775

Pancreas
(Barber 2001)

+2 kg

7 wk fish oil

↓ REE per kg

↑ Appetite/performance

Meta-analysis
(Front Nutr 2021)

↓ CRP

↓ Inflammatory markers in

6 RCTs analyzed

Metabolism stabilized

EPA: ↓ ZAG (lipolysis) • ↓ ubiquitin-proteasome • ↓ IL-6/TNF- α • Sensitizes muscle to insulin • Dose: $\geq 1.5\text{g/day}$ (ESPEN/Colomer)

Neuroprotection: Neuropathy (CIPN)

CIPN affects 30-68% of patients • Only ASCO recommendation: duloxetine (limited benefit)

OR = 0.54

Grade 3-4 CIPN in fish oil
users (SWOG S0221)

SWOG S0221 (PMC6075365):

n=1225 | Breast | Paclitaxel

Animal Model 2025:

Fish oil prevented neuropathy from OXA and PTX. ↓ Spinal microglia and cytokines.

ω-3: one of the few interventions with neuroprotective potential against CIPN — an unmet clinical need

Selective Cytotoxicity: EPA/DHA induce tumor apoptosis but protect normal cells via resolvins and PI3K/Akt

Immunomodulation and Lipid Mediators

Resolvins

(RvE1, RvD1)

- ↓ Neutrophilic infiltration
- ↓ TNF- α and IL-6
- ↑ Apoptotic phagocytosis
- Inflammation resolution

Protectins

(PD1/NPD1)

- Neuroprotection
- ↓ Neuronal apoptosis
- ↑ Pro-survival genes
- DRG protection

Maresins

(MaR1)

- ↑ Tissue regeneration
- ↓ Neuropathic pain
- ↑ M2 macrophages
- ↓ Tumor fibrosis

DHA-WIN Trial (NCT03831178 | PubMed 40651709):

Phase II, double-blind. Neoadjuvant breast. 4.4g DHA/day $\times$ 18 weeks.

DHA modulated oxylipin profile: ↓ pro-inflammatory ω -6, ↑ anti-inflammatory ω -3.

Direct influence on tumor-infiltrating lymphocytes (TILs) — tumor immunomodulation.

Comparison: Ratio $\geq 15:1$ vs. $\leq 3:1$

Domain	Ratio $\geq 15:1$	Ratio $\leq 3:1$
NF- κ B / COX-2	↑↑ Pro-tumor activation	↓↓ Inhibited → pro-apoptotic
Chemosensitivity	↓ (efflux pumps, ↑GSH)	↑ (doxo, cis, 5-FU, OXA)
Response (NSCLC)	25.8%	60.0% (P=0.008)
Progression (CRC)	330 days	593 days (P=0.04)
Body Mass	Loss → cachexia	Stabilization/gain
CIPN	No protection	OR 0.54 grade 3-4
Mortality	+26% total (UK Biobank)	Inverse with mortality
Inflammation	↑↑ TNF- α , IL-6, CRP	↓↓ CRP/IL-6. ↑ Resolvins

Safety and International Guidelines

Safety Profile

0 studies with worse outcomes in the ω -3 group
600mg–3.6g EPA+DHA/day well tolerated
Dose-limiting toxicity similar between groups
SELECTIVE cytotoxicity for tumor cells

ESPEN Guidelines (2021)

Recommends ω -3/fish oil for advanced cancer on CT and at risk of weight loss
Dose: ≥ 1.5 g EPA+DHA/day
ASCO Cachexia (2020): nutritional support as a central component

Suggested protocol: 1.5–3.6g EPA+DHA/day • Start 7-10 days pre-CT • Throughout treatment • Omega-3 Index target $\geq 8\%$

Comparative Scenario: Same Patient

CRC stage III, 58 years, FOLFOX

RATIO 20:1

- Partial response ($\approx 26\%$)
- Progression ~ 330 days
- Loss of 5-8 kg (cachexia)
- CIPN grade 2-3 $\rightarrow$ dose reduction
- CRP/IL-6 chronically elevated
- NF- κ B $\rightarrow$ increasing resistance
- QoL deteriorated

RATIO $\leq 3:1 + \omega-3$

- Potential response up to 60%
- Progression ~ 593 days (+80%)
- Weight stabilized +1-2 kg
- CIPN reduced (OR 0.54)
- $\downarrow$ CRP/IL-6 | $\uparrow$ Resolvins
- NF- κ B suppressed $\rightarrow$ $\uparrow$ sensitivity
- QoL preserved

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Conclusion

Regulating the omega-6:omega-3 balance to $\leq 3:1$ is an evidence-based nutritional intervention with the potential to:

- ✓ Increase tumor sensitivity to chemotherapy
- ✓ Prolong time to disease progression
- ✓ Preserve body mass and combat cachexia
- ✓ Reduce peripheral neuropathy (CIPN)
- ✓ Modulate immune response via resolvins/protectins/maresins

Without interfering with therapeutic efficacy — safety confirmed in multiple RCTs